Employee Performance & Productivity

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An exploratory analysis of 100,000 employee records using Python, MySQL and Power BI. The report focuses on pay, performance, resignation patterns and hire dates in the supplied dataset.

Records	Departments	Job titles	Resigned records
100,000	9	7	10.01%

Summary

Salary varies much more by job title than by the broad demographic groups examined. Average monthly salary ranges from 4,545 for Technicians to 7,799 for Engineers. These figures are reported in dataset units because the original currency documentation is not available.

Employees who stayed and those flagged as resigned have almost the same average satisfaction score: 3.00 and 2.99. Department resignation shares range from 9.56% to 10.54%. These comparisons do not establish why people leave.

Average performance scores are close to 3.00 across all four training-hour groups. That is a descriptive result, not a test of training effectiveness.

The lower annual hire count for 2024 should not be read as a hiring collapse. The extract ends on 3 September. Comparing January–August in both years gives 6,805 records in 2024 versus 6,650 in 2023, an increase of 2.33%.

Scope of the work

The repository contains a Python EDA notebook, 18 MySQL analytical queries with saved CSV outputs, a four-page Power BI dashboard and presentation files. This revised report uses fresh calculations from the bundled CSV. It does not report predictive model results, an automated pipeline or causal effects.

Reading the headline metric: 10.01% is the proportion of records flagged as resigned. It is not an annual turnover rate; the data contains no resignation date.

01 / Data and method

What is included, and what was checked

Source file: Dataset/Extended_Employee_Performance_and_Productivity_Data.csv, stored inside Dataset.zip. The analysis uses the full file rather than a sample.

Data check	Result
Rows and source columns	100,000 rows; 20 columns
Employee identifiers	100,000 unique IDs; no duplicate IDs
Missing values	0 missing cells in the 20 source columns
Hire-date coverage	7 September 2014 to 3 September 2024
Age and tenure	Ages 22–60; tenure values 0–10 years
Record status	89,990 stayed; 10,010 resigned

Approach

Counts, means and proportions were calculated from the source columns. Resignation comparisons use the headcount of the group being compared as their denominator. Salary comparisons are descriptive; no external salary benchmark was applied.

Training groups follow the SQL analysis: under 20, 20–49, 50–79 and 80+ hours. An employee is counted as promoted when Promotions > 0. Work mode is grouped as onsite at 0% remote frequency, fully remote at 100%, and hybrid for values between those endpoints.

Monthly hire records were derived from Hire_Date. The comparison of 2023 and 2024 uses January–August for both years, avoiding incomplete September 2024 and the missing final quarter.

Data limitations

The original source URL, collection method, currency and observation-date documentation are not included in the repository. Hire dates describe when represented employees were hired; they do not prove that the extract contains every hire the organisation made.

Several fields have unusually regular patterns. Each job-title and performance-score combination has one observed salary value, and the five rating groups have nearly equal headcounts. The data is useful for practising analysis, but these results should not be presented as verified findings about a real employer.

The quality checks above establish basic completeness and consistency. They do not establish that the records are authentic, representative or suitable for a real HR decision.

02 / Compensation

Role differences are larger than the aggregate demographic differences

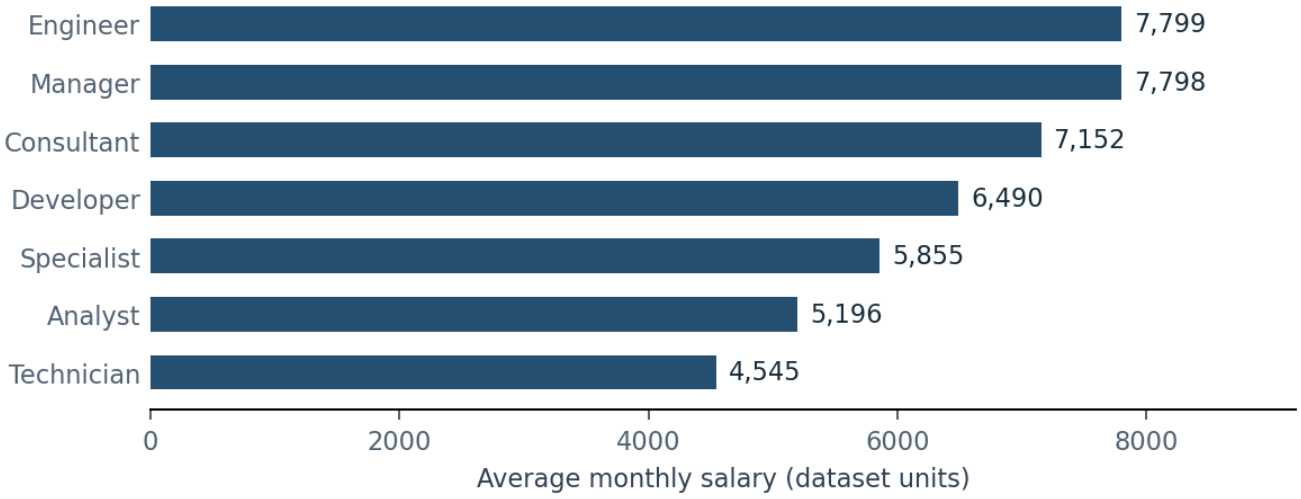


Figure 1. Average monthly salary by job title. All represented records are included.

Engineers average 7,799.32 per month, compared with 4,545.15 for Technicians: a difference of 71.60% relative to the Technician average. This describes different roles; it is not evidence that either role is overpaid or underpaid.

Comparison	Average monthly salary
Female / Male / Other	6,404.99 / 6,400.48 / 6,414.77
Bachelor / Master / PhD	6,398.42 / 6,417.45 / 6,431.55
High School	6,399.36
Tenure groups 0–9 years	Group means range from 6,390.15 to 6,420.38

The broad gender means are close, as are the education-group means. Aggregate means alone cannot establish pay equity or the return to education. Role composition, responsibilities and external market comparisons would be needed for those questions.

Choosing a fair comparison

The existing department-average screening query flags 5,599 top-rated employees as below average. However, departments contain different job titles. Replacing that benchmark with the mean for the same job title and performance score flags none, because salary is identical within each such group in this file. This is a limitation of the dataset, not proof that pay is fair.

Performance score and salary have a Pearson correlation of 0.51. Correlation does not explain the pay-setting process. The 44 records at tenure year 10 are too few for a broad tenure-pay conclusion.

03 / Performance and training

Similar averages do not tell us whether training works

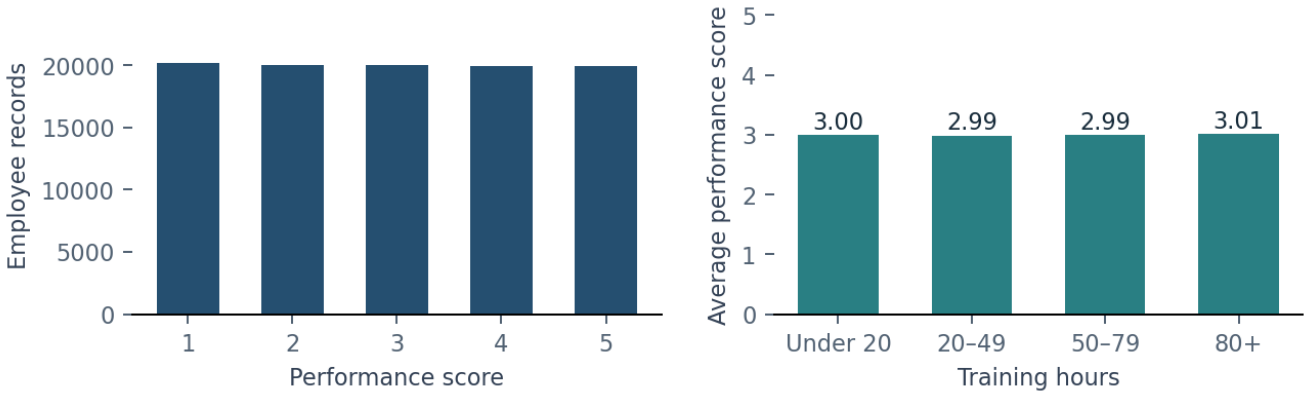


Figure 2. Rating counts and average rating by training-hour group. The rating-average axis spans 0–5 for context.

Training hours	Records	Average score	Promoted (%)
Under 20	20,053	3.00	66.89
20–49	30,086	2.99	66.39
50–79	29,830	2.99	66.67
80+	20,031	3.01	67.05

Performance scores are almost evenly distributed: each score from 1 to 5 accounts for about one-fifth of the records. Average training hours by performance score range from 49.28 to 49.70.

Across training groups, average ratings range from 2.99 to 3.01. The proportion with at least one recorded promotion ranges from 66.39% to 67.05%. The data does not show a clear separation between the groups on these measures.

What would be needed to evaluate training

Training dates, training type, role-specific goals and performance before and after training would make a more useful evaluation possible. Employees are not randomly assigned to training in this analysis, and the timing of promotions is unknown. A cost or return-on-investment estimate is therefore not supported.

Promotion definition: employees with Promotions > 0 divided by all employees in the training group. A zero value is not counted as a promotion. The repository’s current Q17 output uses a different condition and needs to be regenerated.

04 / Resignation patterns

Compare rates within groups, not their contribution to total headcount

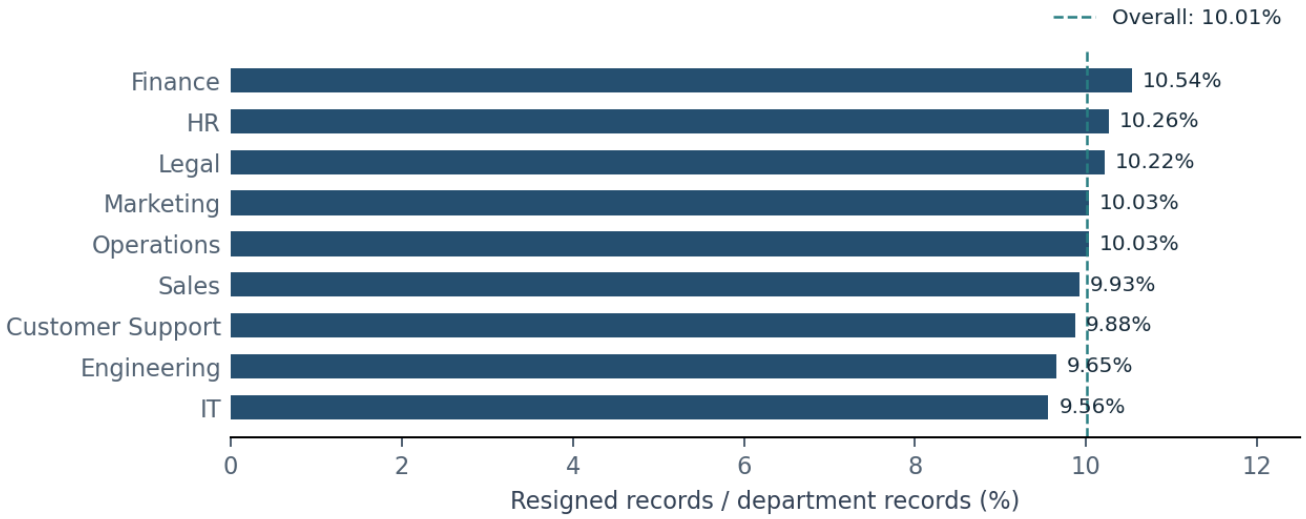


Figure 3. Resigned records as a percentage of each department's records. Bars start at zero.

Finance has 1,180 resigned records out of 11,200, or 10.54%. IT has 1,064 out of 11,131, or 9.56%. The difference is 0.98 percentage points. These descriptive differences alone are not enough to prioritise a retention intervention.

Work mode	Records	Resigned	Share (%)
Onsite	19,837	1,881	9.48
Hybrid	60,114	6,091	10.13
Fully remote	20,049	2,038	10.17

Onsite and fully remote groups differ by 0.68 percentage points in the observed resignation share. The groups were not matched for role, tenure or other characteristics, so this is not an estimate of the effect of remote work.

Measure	Stayed	Resigned
Average satisfaction score	3.00	2.99
Average overtime hours	14.51	14.58

Group averages are close for both satisfaction and overtime. Overtime is compared using means, not totals: there are roughly nine times as many stayed records as resigned records. Totals would mainly reflect the different group sizes.

No exit date or resignation reason is available. This report therefore does not estimate annual turnover, time-to-exit, future resignation risk or the cost of attrition.

05 / Hiring records

Use the same calendar months for year-on-year comparisons

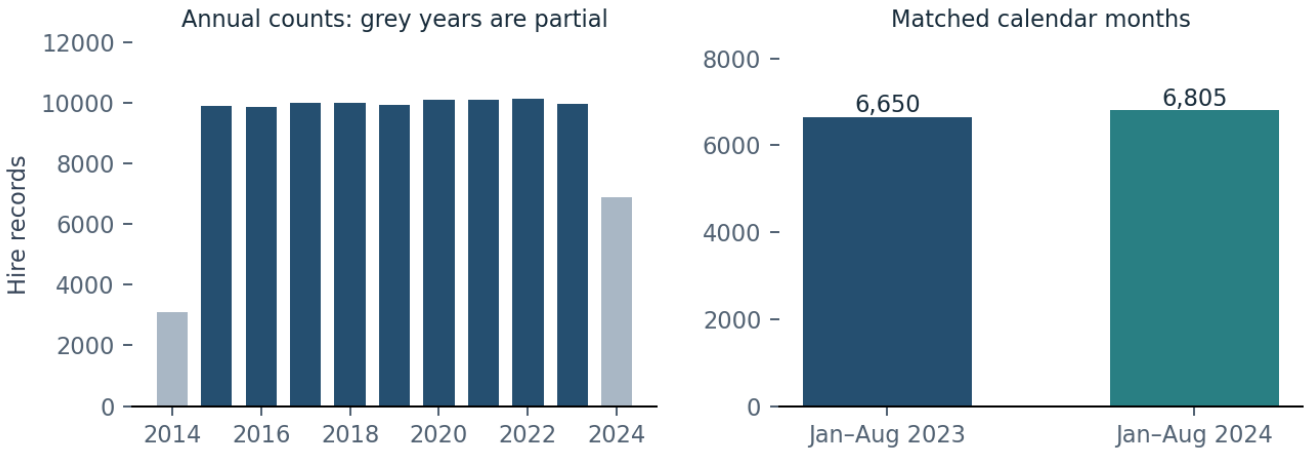


Figure 4. Annual hire records and a matched January–August comparison. Annual 2014 and 2024 counts have incomplete calendar coverage.

Period	2023	2024	Change
Full calendar-year records	9,973	6,890*	Not comparable
January–August records	6,650	6,805	+155 / +2.33%

*The 2024 figure only includes dates through 3 September. The 2014 extract starts on 7 September. Neither is a complete calendar year.

The raw annual comparison would show a 30.91% reduction from 2023 to 2024, but the observation periods differ. On a matched January–August basis, 2024 has 155 more represented hire records than 2023.

What this supports

The apparent drop in the unadjusted 2024 chart is not sufficient evidence of reduced hiring. The matched-period result is a more appropriate descriptive comparison for the available records.

What this does not support

The file cannot establish company growth, restructuring or a change in recruitment policy. That would require complete hiring and exit histories, a known observation date and confirmation that all relevant employees are represented.

This calculation uses the month of Hire_Date. It does not annualise a few days of September or extrapolate a full-year 2024 total.

06 / Definitions and follow-up

Keep the calculations and repository outputs aligned

Metric	Definition used in this report
Resignation share	Resigned records in a group / all records in that group × 100
Promotion share	Records with Promotions > 0 / all records in the group × 100
Pay-gap percentage	(Male mean salary − Female mean salary) / Male mean salary × 100, when using the Q9 convention
Education salary index	Group mean salary / High School mean salary; an index, not a percentage
Salary benchmark	Compare like-for-like roles before labelling an individual underpaid

Corrections relative to the saved SQL outputs

Q6 and Q7 currently divide segment resignations by total company headcount. That measures a contribution to total records, not a within-group resignation share. Q9 reports a ratio under a percentage label, and Q11 reports an index under a percentage label.

Q13 uses company headcount for a department-level burnout screen. Using each department's headcount gives a flagged share of 3.37%–3.88%. The rule is more than 50 work hours per week, more than 20 overtime hours and satisfaction below 2.5. It is a heuristic screen, not a validated health measure.

Q17 counts Promotions >= 0. This revised report uses Promotions > 0. The report figures have been recalculated directly from the CSV; the existing SQL scripts and saved outputs have not been changed by producing this document.

Recommended next steps

1. Fix the MySQL loader's column naming and derived resignation status, then rerun the affected queries and replace the saved outputs.

2. Check Power BI measures against the corrected results. Label partial hiring years and distinguish average overtime from total overtime.

3. Add the original dataset source and field definitions. For any real retention study, collect exit dates and reasons before building time-based metrics or recommending policy changes.

Calculation basis: Dataset.zip from repository commit e851ace. All figures use the supplied extract; they are not current workforce statistics. No predictive models, statistical significance tests or automated deployment claims are included in this report.